



USERS GUIDE

Radiant QuikRAM

HDR and radioactive-materials management with built-in TG-43.

QuikRAM
RADIANT MEDICAL PHYSICS CONSULTING

InventorySource ModelsWell ChambersHDR Source ExchangesTG-43 CalcDecay TableReportsLOCAL-FIRST - SQLITE

Source Inventory

2 sources tracked — May 23, 2026. Active sources due for HDR source exchange within 30 days are flagged below.

0
OK

1
EXCHANGE DUE

0
OVERDUE

0
NO DATE

1
INACTIVE

+ Add Source

+ Add Source Model

+ Add Well Chamber

+ Record HDR Source Exchange

Printable Inventory

Status	Type	Model / Isotope	S/N	Location	Cal Date	Initial S_K (U)	% Remaining	Exchange Due
Exchange Due Soon	HDR Source	Nucletron mHDR-v2 (Ir-192)	24-01-3914-001-021226-12708-26	HDR Vault	2026-02-12	51722.000	39.1%	2026-06-10
Removed	HDR Source	Nucletron mHDR-v2 (Ir-192)	24-01-3670-001-111725-12377-76	HDR Vault	2025-11-17	50374.000	17.3%	2026-02-17

Radiant QuikRAM — a tool of Radiant Medical Physics Consulting LLC. Runs 100% locally; no data leaves this machine.
Source records, chamber-calibration values, and TG-43 calculations must be independently verified by a Qualified Medical Physicist before clinical use.

What QuikRAM is.

Radiant QuikRAM is HDR and radioactive-materials management for radiation oncology — source inventory with automatic isotope decay correction, the full HDR Source Exchange QA worksheet, well-chamber inventory with ADCL calibration values, a TG-43 line-source dose calculator, and printable date / AKS / activity decay tables — all in one local web app.

Who it's for

- HDR brachytherapy programs.
- Radiation safety officers managing sealed sources.
- Clinical physicists doing TG-43 reference checks.

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SOURCE INVENTORY — DECAY-CORRECTED AKS TODAY.

Access and your source inventory.

Access

QuikRAM is deployed at quikram.radiant-mpc.com. Open the URL in any modern browser.

Add a source

1. From the dashboard, click *+ New source*.
2. Pick the isotope (Ir-192, Co-60, Cs-137, I-125, Pd-103, Y-90, etc.). QuikRAM uses the published half-life for decay corrections.
3. Enter the calibration date, the calibration air-kerma strength (or activity), and the certificate number.
4. Save. The source dashboard shows the AKS / activity decay-corrected to today.

QuikRAM
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InventorySource ModelsWell ChambersHDR Source ExchangesTG-43 CalcDecay TableReportsLOCAL-FIRST · SQLite

Inventory / 24-01-3814-001-021226-12708-26

24-01-3814-001-021226-12708-26 — Nucletron mHDR-v2

Exchange Due Soon · 18 d

HDR Source · Ir-192 · HDR Vault

Details

Type	HDR Source
Model	Nucletron mHDR-v2 — Ir-192
Serial number	24-01-3814-001-021226-12708-26
Location	HDR Vault
Status	Active (in service)
Cal date	2026-02-12
Initial S_K (U)	51722.0000
Current S_K (2026-05-23)	20227.4591 U — 39.11% remaining
Install date	2026-03-10
HDR source exchange due	2026-06-10
Removed / disposed	— / —
Notes	Q1-2026 source, AMC 72-3814 / in-house ID.

Edit source

TG-43 calc →

Delete

SOURCE DETAIL — DECAY CORRECTION TO TODAY.

HDR Source Exchange QA.

The Source Exchange workflow walks you through every section of the quarterly QA in order, with auto-computed metrics where the math is mechanical.

- 1. **ODH acceptance checks** on the new source.
- 2. **Chamber constancy** on the prior source (before removal).
- 3. **New source calibration** — well-chamber reading, correction factors, calculated AKS, comparison to vendor cert.
- 4. **Timer accuracy, reproducibility, and linearity** with an auto-computed R-squared plot.
- 5. **Housing survey** and **source-positioning film** upload.

At the end, click *Generate report* for a complete QA PDF ready for the source binder.

QuikRAM

RADIANT MEDICAL PHYSICS CONSULTING

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LOCAL-FIRST · SQLite

HDR Source Exchanges / 2026-03-10

HDR Source Exchange QA

2026-03-10 · Julie Walls, M.S., DABR · OUT: 24-01-3670-001-111725-12377-76 · IN: 24-01-3814-001-021226-12708-26

Summary PASS

Edit

Upload positioning film

Printable QA report

Decay table (new source)

Delete

Summary

Test	Measured	Limit	Verdict
Chamber Constancy Check	-0.35%	±2.0%	PASS
New Source Calibration	+0.51%	±5.0%	PASS
Timer Accuracy	-0.14 s	±0.60 s	PASS
Timer Reproducibility	+0.01%	2.0%	PASS
Timer Linearity (worst)	+0.02%	2.0%	PASS
Source Positioning (see film)	0.0 mm	1.0 mm	PASS
Source Housing Survey	0.08 mR/hr	0.30 mR/hr @ 1 m	PASS

Other items: New source entered in RadCalc / Point Dose Check — YES; Activity table printed and posted — YES.

ODH acceptance checks

HDR SOURCE EXCHANGE QA — SECTIONED WORKSHEET.

Well-chamber inventory + TG-43 calculator.

Well-chamber inventory

Each well chamber is a tracked asset with manufacturer, serial number, ADCL calibration coefficient, calibration date, and the cross-calibration history. The dashboard shows when a chamber's ADCL is due.

TG-43 line-source calculator

An independent dose-rate reference check. Pick the source, the distance, and the off-axis angle; QuikRAM applies the published $g(r)$, $F(r, \theta)$, and anisotropy data and returns the dose-rate at point. Useful as a second check on planning-system output, especially during commissioning.

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Well Chambers / Edit

Edit well chamber

Calibration values come from the ADCL cert. The chamber is then selectable on the HDR source exchange form for verification measurements.

Calibration status today · -42 d

Overdue

IDENTITY

ManufacturerModel

Standard ImagingHDR1000+

Serial numberLocation

A200788HDR Vault

Status

Active (in service)

CALIBRATION VALUES (FROM ADCL CERT)

Cal isotopeN_{cal} calibration coefficientN_{cal} units

Iridium-192467800.0Cy·m²/hr/A

Cal dateCal due dateCert / ADCL #

04/11/202404/11/2026ADCL-2024-A200788

OPERATING CONDITIONS

WELL-CHAMBER DETAIL — ADCL COEFFICIENT AND HISTORY.

Decay tables and binder reports.

Date / AKS / activity decay tables

Pick a source and a horizon (30, 60, 90, 180 days) and QuikRAM prints a day-by-day decay table — the table the source binder needs and the table that backs the day's planning calculations.

Source-exchange report

The exchange workflow rolls every section into a Radiant-branded PDF: acceptance checks, constancy, calibration, timer / reproducibility / linearity, housing survey, and film image, with the reviewer signature block on the last page.

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Inventory / Decay Table

Source Date / AKS / Activity Decay Table

Daily decay-corrected air-kerma strength (S_K) and activity for any source, starting from any date for N days. Pick a source, set a start date and number of days, then print for the source binder (pages 24-25 of the QA worksheet).

Source

Start date defaults to source cal date

Days max 1000

24-01-3814-001-021226-12708-26 — Nucletron m

mm/dd/yyyy

180

Generate table

Reset

24-01-3814-001-021226-12708-26 — Nucletron mHDR-v2 (Ir-192)

Calibration date	2026-02-12
Calibrated S_K (U)	51722.0000
Calibrated AKR (mCy/hr @ 1 m)	51.7220
Calibrated activity (Ci)	12.7080
Half-life	73.830 days
Window	2026-02-12 — 2026-08-10 (180 days)

Print / Save as PDF

Date	Days	AKS (U)	AKR (mCy/hr @ 1 m)	Activity (Ci)
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DECAY TABLE — DATE, AKS, ACTIVITY COLUMNS.

License, safety, and how to reach us.

License terms

QuikRAM is licensed per HDR program. The deployed instance at quikram.radiant-mpc.com is provided as a hosted evaluation; a self-hosted on-prem build is available on request.

Evaluation status

QuikRAM is an unvalidated analysis aid. Every TG-43 calculation and every source-exchange QA conclusion must be independently reviewed by a Qualified Medical Physicist before clinical use.

Contact

Live URL: <https://quikram.radiant-mpc.com/>

Email: info@Radiant-MPC.com

Website: radiant-mpc.com

Evaluation / decision-support only. Every Radiant tool is an aid to a Qualified Medical Physicist. All outputs must be independently reviewed before any clinical use. Do not transmit PHI through evaluation builds.